

Exploring Emotion-related Individual Differences in Adolescents' Daily Lives: An Analysis Using Large Language Models

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Abstract: Diary methods can help us understand how adolescents recognize and manage emotions in their daily lives, which aids in optimizing social-emotional learning practices. This exploratory study invited 30 students to write their daily entries. These unstructured texts were then automatically coded with the help of large language models, which performed comparably to human raters. Importantly, two indicators (semantic similarity and emotional event alignment) were used to assess how an individual's feelings in specific contexts align with the common understanding of those contexts. We found positive associations between diary-derived indicators for positive entries and performance on emotion perception tasks, while students with lower semantic similarity for negative entries showed higher performance on the emotion regulation task, indicating self-regulation in negative contexts. The study highlighted the potential of using large language models to characterize students' daily emotions and preliminarily demonstrated how individual differences manifest in adolescents' emotional lives.

Keywords: adolescents; emotion; individual difference; diary methods; large language models

Introduction

Adolescence involves substantial changes in emotional functioning, with vulnerability to emotional problems (Silvers, 2022). To design social-emotional learning practices during this critical period, it is crucial to characterize how adolescents recognize and manage emotions, especially in their daily lives. Two methods are commonly used to characterize these emotion-related individual differences (Austin, 2004). The first method relies on performance data from emotion-related tasks, in which individuals' responses are compared to predetermined answers to assess their ability (Floman et al., 2023; Wang et al., 2021). Another approach utilizes self-report questionnaires, asking people to reflect on and report their perceptions (Petrides et al., 2007). While both methods have offered valuable insights for understanding adolescents' emotion-related individual differences, they typically offer a holistic overview of these differences, making it difficult to cover more nuanced information about what happens in adolescents' daily lives.

The diary method is an important data collection tool for tracking daily entries (Scott, 2022). It is also suggested that it holds the potential to reveal emotional patterns related to individual differences. For instance, researchers have used diary methods to track the dynamic process of an individual's emotions during the pandemic and found that higher emotional intelligence predicted lower intensity of fear, anxiety, and sadness (Moroñ & Biolik-Moroñ, 2021). Another study specifically explored the effectiveness of electronic diaries in youth populations and suggested that such diaries could provide important information on emotional functioning (Suveg et al., 2010). However, much of the existing research has focused on the intensity and frequency of emotions, sometimes overlooking the contextual information related to the emotions. As widely acknowledged in social psychology, emotions are inherently context-dependent (Greenaway et al., 2018). It is necessary to anchor the emotion within the specific context that triggers it to achieve a thorough understanding (Dejonckheere et al., 2019; Lapate & Heller, 2020). Therefore, we aim to explore daily emotional patterns by analyzing the contextual information of adolescents' daily entries in conjunction with the corresponding emotions. Nevertheless, reporting contextual information in diaries may generate unstructured text, which requires manual coding—a process that is both time-consuming and resource-intensive.

Large language models (LLMs) provide an efficient approach for processing unstructured data (Gilardi et al., 2023), e.g., sentiment analysis of written responses (Hur et al., 2024; Rathje et al., 2024). The present study aims to further examine the feasibility of using LLMs to explore emotion-related individual differences from adolescents' diary texts. Importantly, rather than separately categorizing contexts and emotions (e.g., categorizing contexts into academic, family, and school relationships, and categorizing emotions into joy, sadness, etc.), analyzing each context together with its corresponding emotions allows for a more nuanced understanding of the unique characteristics of each specific entry. This will also provide us with an opportunity to characterize individualized emotional responses that may be more accurately reflected in subtle contextual information.

Therefore, inspired by the semantic similarity analysis commonly used in natural language processing (Araque et al., 2019), this study employs an "emotional event alignment" metric to integrate LLM-rated emotion scores by computing the distance between context and emotion. We expect that this metric can show how individuals' actual emotions in a specific context (quantified by the emotional ratings of the subjective feelings) differ from the common understanding for this context (quantified by the emotion scores of the context if we consider the LLM-rated scores as a generalized emotional reflection of the context). As a supplement, we also computed the semantic similarity metric to directly capture semantic distances between contextual information and corresponding emotions based on their embeddings, as the embeddings contain rich human knowledge (Grand et al., 2022).

By calculating semantic similarity and emotional event alignment, we can integrate the analyses of context and emotions for each entry. We hypothesize that these indicators may be associated with individual differences in emotion-related tasks, as they quantitatively describe adolescents' emotional patterns that happen in daily life. For instance, when an individual's actual emotional response is more consistent with the emotions expected to be triggered in an entry, it may indicate the person can more accurately perceive the emotions involved. However, the relationship could be complex. For example, the high consistency with consensus does not necessarily mean a positive outcome, especially in the case of negative entries. It may suggest that the individual is unable to effectively regulate and manage emotions, making them more susceptible to being influenced by negative entries. Nevertheless, as this study is still in its preliminary and exploratory phase, we seek to address two questions focusing on early-stage insights here: 1) Can LLMs rate the emotions expressed in students' diary entries? 2) Is the diary-derived indicator, which links the context of specific entries to corresponding feelings, associated with adolescents' differences in recognizing and managing emotions?

Methods

Participants and data collection

Thirty first-year middle school students aged 15 to 16 years participated in the diary-based data collection and received small non-monetary gifts (e.g., stationery) as compensation. During the data collection, the content of the tasks was first explained to the students. Then, students filled out online emotion diaries for two consecutive weeks ($M = 4.2$ days, $SD = 3.4$ days). Each day at 5 PM (after school), students received the diary link on their mobile phones. They were asked to write about one negative entry and one positive entry and to describe the emotions they felt in each. For example, a student shared a positive entry, where the context was that their class was practicing singing together. The student reported feeling happy, satisfied, and connected to their classmates because of everyone's improvement. Students could submit their diary entries until 7 AM the next day. Any entries after that were not included. Students reported a total of 126 emotion daily entries. Before analysis, all data were anonymized, and any potentially identifiable information, such as the names of students mentioned in the entries, was masked. Students also participated in emotion perception and emotion regulation tasks derived from Wang et al. (2021). In the emotion perception task, students needed to recognize emotions from faces, while in the emotion regulation task, students needed to choose the worst/best strategies to achieve emotion regulation goals. Their emotion-related individual differences were quantified based on the corresponding task performances, which were converted into z-scores. The emotion understanding task was not included due to technical issues.

Text analysis based on LLMs

To integrate the context information and emotions, this study uses two indicators for text analysis. The first is emotional event alignment. To compute this measure, we use a prompt to guide ChatGPT 4omini in rating the emotional aspect for the context and corresponding subjective feelings for each daily entry across 12 dimensions (6 positive and 6 negative dimensions): 'sadness,' 'anger,' 'disappointment,' 'guilt,' 'anxiety,' 'stress,' 'joy,' 'excitement,' 'satisfaction,' 'curiosity,' 'inspiration,' and 'fascination'. This yields an emotional rating vector for each context and a corresponding vector for students' feelings. The first vector can be regarded as a common emotional pattern triggered by the entry, assuming that GPT's scoring reflects human emotional consensus. The second vector, on the other hand, quantifies students' emotional responses. By calculating the cosine similarity between the two vectors, we can determine how each student's emotional response to a specific entry differs from the GPT-rated emotional response.

$$EmotionalEventAlignment = \frac{Rating(Context) \cdot Rating(Emotion)}{||Rating(Context)|| \cdot ||Rating(Emotion)||} \quad (1)$$

The LLM prompt used is "Please analyze the emotions reflected in the following text on the '{dimension}' dimension, and make a numerical rating from 0 to 6, (0 is none, 6 is very high) (no explanation

needed):\n\n' {text}'\n\n Emotional Dimension: {dimension}\n Rating (0 to 6): " The design of the prompt draws on the wording from classical self-assessment questionnaires, aligning the scoring between humans and LLMs.

To evaluate the accuracy of GPT's emotional ratings, two independent raters with no prior knowledge of the study were invited to rate 120 randomly selected texts from the daily entries using the same prompt. These included 60 context texts (30 positive, 30 negative) and 60 feeling texts (30 positive, 30 negative). Before being provided to the raters, the entries were cleaned of any potentially identifiable personal information. Then, the correlation between human-rater-based scores and GPT-based scores was calculated for each dimension.

The second indicator, semantic similarity, is computed as the distance between embeddings of context and feelings. Specifically, we use OpenAI's API (Version 1.52.2, Python) to get the embedding of texts describing context and feelings, and then compute the cosine similarity between the two embeddings:

$$\text{SemanticSimilarity} = \frac{\text{Embedding}(\text{Context}) \cdot \text{Embedding}(\text{Emotion})}{\|\text{Embedding}(\text{Context})\| \cdot \|\text{Embedding}(\text{Emotion})\|} \quad (2)$$

The relationship between diary indicators and emotion-related individual differences

To explore the association between individual emotional patterns reflected in diary-related indicators and their emotion-related individual differences, we constructed a regression model as follows:

$$\text{DiaryIndicators}_{ij} = \beta_0 + \beta_1 \text{EmotionIndividualDifference}_i + \mu_{0i} + \epsilon_{ij} \quad (3)$$

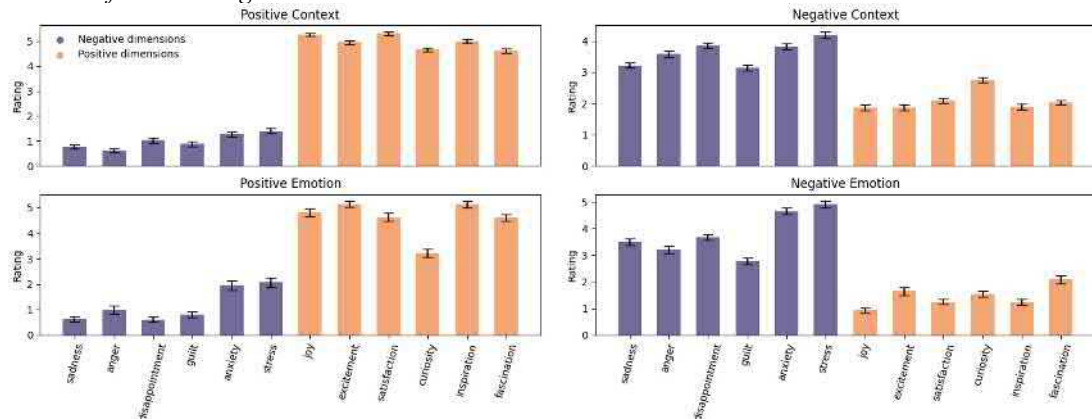
where $\text{DiaryIndicators}_{ij}$ denotes the emotional event alignment (or semantic similarity) of student i in entry j , and $\text{EmotionIndividualDifference}_i$ denotes one of the emotion-related individual differences (i.e., task performances for emotion perception or emotion regulation) of student i . Considering that each student reported more than one entry, the variance of the residuals is composed of two parts: within-person variance and between-person variance. Thus, the models are set up with clustered standard errors μ_{0i} .

Results and discussion

First, we visualized the LLM ratings for context texts (positive and negative), as well as the subjective feeling texts (positive and negative) on the 12 emotional dimensions. As shown in Figure 1, the LLM ratings indeed assign higher scores to the emotional dimensions that correspond to the respective emotion categories of entries.

Figure 1

Bar Plots of LLM Ratings on the 12 Emotional Dimensions



Here, the positive context (or feeling) indicated that the context was reported in a positive entry. The same goes for negative ones. The errorbar represents S.E.M.

Then, we calculated the correlation between LLM ratings and human ratings for further validation. As shown in Table 1, the scores from the LLM and human raters showed significant positive correlations, with high correlation coefficients across most of the emotional dimensions. This demonstrates the effectiveness of the LLM in automatic text coding on emotional dimensions, consistent with (Amin et al., 2024; Niu et al., 2024). However, it is important to note the relatively lower correlation coefficients (<0.4) for 'guilt,' 'curiosity,' and 'fascination'. A possible explanation could be that these dimensions lack clear linguistic indicators, making them more difficult to accurately infer from text, while emotions like 'sadness' are often expressed through more explicit cues. Therefore, the consistency between human and LLM ratings in these dimensions is relatively weaker, suggesting the need to further improve accuracy.

Table 1

The Correlation between LLM Ratings and Human Ratings

	Sadness	Anger	Disappointment	Guilt	Anxiety	Stress
Context -H1	0.829 ***	0.777 ***	0.694 ***	0.594 ***	0.779 ***	0.775 ***
Context -H2	0.857 ***	0.720 ***	0.838 ***	0.660 ***	0.840 ***	0.844 ***
Emotion-H1	0.712 ***	0.467 ***	0.497 ***	0.259 *	0.881 ***	0.864 ***
Emotion-H2	0.852 ***	0.584 ***	0.792 ***	0.367 **	0.853 ***	0.927 ***
	Joy	Excitement	Satisfaction	Curiosity	Inspiration	Fascination
Context -H1	0.924 ***	0.911 ***	0.948 ***	0.382 **	0.870 ***	0.885 ***
Context -H2	0.897 ***	0.825 ***	0.917 ***	0.417 **	0.780 ***	0.752 ***
Emotion-H1	0.900 ***	0.829 ***	0.869 ***	0.501 ***	0.741 ***	0.377 **
Emotion-H2	0.852 ***	0.792 ***	0.870 ***	0.582 ***	0.652 ***	0.446 ***

Note: H1: human rater 1, H2: human rater 2, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Next, as shown in Table 2, we found significant associations between emotion-related individual differences and diary-derived indicators. Students' performance in the emotion perception task was positively correlated with their emotional event alignment scores for positive events, which was consistent with our hypothesis, i.e., an individual's feelings aligning with the expected emotions of an entry may indicate better perception and understanding of its emotional content. This finding was replicated in the semantic similarity scores with a lower coefficient. We also found that students with higher emotion regulation performance showed a negative correlation with their semantic similarity scores for negative entries. This might suggest that they could manage or downplay their negative emotional responses, leading to a reactivity that is less aligned with the typical patterns associated with negative entries. The results are consistent with previous findings that reported individuals with high emotion regulation abilities have decreased tendencies to experience negative feelings (Mikolajczak et al., 2008). One possible explanation is that emotion regulation processes might influence students more broadly, affecting not only their emotional aspect of subjective feelings but also their cognitive and behavioral responses. For instance, a well-regulated student might view the high-pressure context as a challenge rather than a threat, which can lead them to approach the context with a more proactive mindset, influencing their overall perception and response to the situation. This richer information beyond emotions can be captured by embeddings (Grand et al., 2022), unlike emotional event alignment, which focuses solely on the emotional dimension of the text. However, it is important to note that emotional event alignment offers better interpretability, because of its clearer psychological meaning over semantic similarity. Also, the difference between positive and negative entries underscores the necessity of considering entries with different valences when exploring emotion-related individual differences in daily situations.

Table 2

The Association between Emotion-Related Individual Differences and Diary-Derived Indicators

	Emotional event alignment		Semantic similarity	
	On positive event	On negative event	On positive event	On negative event
Emotion perception	0.307 *** (0.086)	0.027 (0.090)	0.236 * (0.109)	-0.072 (0.203)
Emotion regulation	-0.098 (0.089)	-0.133 (0.089)	0.117 (0.106)	-0.347 * (0.168)

Note: all coefficients are standardized. Standard error in parentheses. * $p < 0.05$, *** $p < 0.001$

Conclusion and future direction

In summary, the present study highlights the potential of using LLMs to characterize students' tendencies when dealing with everyday entries from diary texts and preliminarily demonstrates how individual differences manifest in adolescents' daily lives. Further efforts are needed to inform the development of tailored SEL programs that offer more personalized and effective interventions by detecting the emotional patterns unique to each student. In addition, it is important to be cautious about the hallucination issue of LLMs, which may benefit from manual validation. Moreover, the preliminary and exploratory nature of this study should be taken into consideration given the relatively small sample size. Additionally, note that the Cronbach's alpha values of the task performances are marginally acceptable (around 0.7). Further investigation is needed to better capture emotion-related individual differences.

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